

Unit 3: Measures of Central Tendency

Concept – To express raw data in the form of frequency distribution is the first step towards condensation of data in compact form. In this form the data becomes easy to understand and many features of the data such as pattern of variation, position of concentration symmetry of frequency distribution etc. are clear

Now suppose we have two frequency distributions showing marks of students of two different colleges and question is asked that which college shows better performance? To answer this question with the help of only frequency distribution is misleading because one can't assimilate the frequency distribution at a time. Therefore to make proper comparison, there is need to condense a frequency distribution into a single figure which tells whole story about data. This single figure is a representative figure and gives complete picture of the phenomenon under study.

It should be natural to expect that the representative figure should be more or less close to most of the figures in the data. Hence it lies more or less at the central position in the data. This tendency of representative figure is called central tendency. The measures used to measure this central tendency are known as measures of central tendency.

Requirements of Good measures of central tendency –

- i) It should be simple to understand and easy to calculate.
- ii) It should be clearly and rigidly defined.
- iii) It should be based on all observations.
- iv) It should not be affected by extreme observations.
- v) It should be capable of further algebraic treatments.
- vi) It should have sample stability.

Objects of measures of central tendency

- i) To get single representative figure
- ii) To compare one set of data with other set of data

We have to study following measures of central tendency

- i) Arithmetic mean ii) Median iii) Mode

1) Arithmetic mean (A. M.). Def.- Arithmetic mean is defined as sum of all observations divided

by total number of observations.

i.e. let $X_1, X_2, X_3, \dots, X_n$ be the n observations on variable X , it's A.M. is denoted by $\bar{X}$ and given by $\bar{X} = [X_1 + X_2 + X_3 + \dots + X_n] / n$

$$= \frac{\sum_{i=1}^n X_i}{n} = \frac{\sum X}{n}$$

Computations of A.M.---

Case-I) Individual data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations on variable X , it's A.M. is calculated by formula

$$\text{A.M.}(X) = \frac{\sum X}{n}$$

Ex.1) The marks obtained by a student in 6 subject are 67, 56, 59, 60, 72, 48. find A.M. of marks.

Ex.2) The mean of 8 observations 23, 33, 25, 16, a, 28, 37, 35 is 27 find the value of 'a'.

Case-II) Discrete frequency distribution- let $X_1, X_2, X_3, \dots, X_n$ be the observations

having frequencies $f_1, f_2, f_3, \dots, f_n$ respectively, it's A.M. is calculated by formula

$$\text{A.M.}(\bar{X}) = \frac{\sum fX}{N}, \text{ Where } N = \sum f$$

Ex.1) Find mean wages of workers from the following data.

Wage (Rs.): 20 25 30 40 43 55 58 64

No. of workers: 3 8 12 17 15 14 9 4

Ex.2) Find mean for the data given below

X: 5 7 8 9 10 12 13 15 16

f: 2 6 9 15 11 10 7 4 1

Case-III) Continuous frequency distribution- let classes for the values of variable and their respective frequencies are given, it's A.M. is calculated by formula

$$\text{A.M.}(\bar{X}) = \frac{\sum fm}{N}, \text{ Where } N = \sum f, m = \text{midpoints of classes}$$

Ex.1) Find mean for the data given below.

Class: 15-25 25-35 35-45 45-55 55-65 65-75 75-85

Freq.: 6 12 16 21 13 7 3

Ex.2) Following is the freq. distribution showing height of students of B.A.-I class, find A.M. of height.

Height (in c.m.): 145-150 150-155 155-160 160-165 165-170 170-175

No. of students : 11 17 24 20 14 9

Properties of A.M.-

1) The product of A.M. and total number of observations is equal to sum of all observations.

$$\text{i.e. } n \cdot \bar{X} = \sum X$$

2) The algebraic sum of deviations of observations taken from it's A.M. is zero.

$$\text{i.e. } \sum (X_i - \bar{X}) = 0$$

3) Sum of squares of deviations of observations is minimum if it is taken from it's A.M.

$$\text{i.e. } \sum (X_i - \bar{X})^2 \leq \sum (X_i - A)^2, \text{ where } A \text{ is any constant other than } \bar{X}$$

4) If $\bar{X}_1$ and $\bar{X}_2$ be the arithmetic means of two groups having sizes n_1 and n_2 respectively, then

A.M. of combined group is given by

$$\text{Combined A.M.}(\bar{X}) = (n_1 \bar{X}_1 + n_2 \bar{X}_2) / (n_1 + n_2)$$

5) Effect of change of origin and scale on A.M.-

let $X_1, X_2, X_3, \dots, X_n$ be the 'n' observations

$$\therefore \text{By definition } \bar{X} = \frac{\sum X}{n}$$

$$\text{let. us define } d = \frac{X - A}{B}, \text{ where } A \text{ \& } B \text{ are constants}$$

$$\therefore \text{By definition } \bar{d} = \frac{\sum d}{n}$$

$$\therefore \bar{d} = \frac{\sum \left(\frac{X - A}{B} \right)}{n}$$

$$\therefore \bar{X} = B \bar{d} + A$$

This implies that A.M. is depends up on change of origin and scale both.

Ex.1) Find A.M. for the following data.

Class :	125-150	150-175	175-200	200-225	225-250	250-275	275-300
Freq. :	12	21	28	35	31	23	11

Ex.2) The mean salary of 77 workers in a certain factory is Rs. 1560. The mean salary of 32 of them is Rs. 1500 and that of next 25 of remaining is Rs. 1640. What is mean salary of remaining 20 workers.

Ex.3) The average monthly sales for the first 11 months of a year of a salesman was Rs. 12000, but due to his illness during last month the average monthly sales for the whole year came down to Rs. 11375. find sale in last month.

Ex.4) The mean monthly salary paid to all employees in a certain factory was Rs. 6000. the monthly salary paid to male and female employees were Rs. 6200 and Rs. 5200 respectively.

Obtain the ratio of male and female employees in the factory.

Ex.5) The mean of 100 items is found to be 30. Later on it was found that two observations 32 and 12 are wrongly taken as 23 and 11. find correct mean.

Weighted A.M.-

The A.M. gives equal importance to each of the items in the data, but there are many problems in which it not true e.g. importance of wheat in our meals is not same as that of salt. Also importance of edible oil and kerosene is different. Thus wheat, rice, house rent, fuel etc. are more important than tea, sweetmeats, cosmetics etc. In this case simple A.M. is not proper representative. So we want to find an average which takes into consideration the relative importance of these items. One such average is weighted A.M. The figures which represents the relative importance of an item is called weight of that item in the data.

Def.- Weighted A.M. is defined as sum of products of weights and values divided by total weights.

i.e. if $w_1, w_2, w_3, \dots, w_n$ are the weights attached to the values $X_1, X_2, X_3, \dots, X_n$

respectively, then weighted A.M is given by

$$\text{weighted A.M.} = \frac{\sum wX}{\sum w}$$

Note- The difference between frequency and weight is that frequency is always in integer but weight may be integer or fraction.

Ex. Calculate simple and weighted A.M. of price per ton of coal from the following data.

month	Price per ton (Rs.)	Tons purchased
Jan	42.5	25
Feb	51.25	30
Mar	50.0	40.5
April	44.25	50
May	54.0	10.5
June	52.6	45.5

Merits and Demerits of A.M.

- Merits:-**
- i) it is simple to understand and easy to calculate.
 - ii) it is clearly and rigidly defined.
 - iii) it is based on all observations.
 - iv) it is capable for further algebraic treatments.
 - v) it has sampling stability.

- Demerits:-**
- i) it is affected by extreme observations.
 - ii) it is sometimes not an actual value in the data.
 - iii) it cannot be determined graphically.
 - iv) it is not suitable for open-end classes.
 - v) it is not applicable for qualitative data.

2) Median. Def- Median is defined as size of item which lies at centre, when data is arranged in ascending or descending order of magnitude.

$$\text{i.e. Me} = \text{size of } \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item}$$

Note- i) No. of observations having value less than median are equal to No. of observations having

value greater than median

ii) Median is also known as positional average because it refers position (place) in the given data .

iii) Median = size of $\left(\frac{n+1}{2} \right)^{\text{th}}$ item , if n is odd

$$= \{ \text{size of } (n/2)^{\text{th}} \text{ item} + \text{size of } (n/2 + 1)^{\text{th}} \text{ item} \} / 2 , \quad \text{if n is even}$$

Computations of Median.---

Case-I) Individual data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations on variable X, it's Median. is obtained as follows

i) Arrange the data in ascending order

ii) $\text{Me} = \text{size of } \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item}$

Ex.1) Find median for the data 17, 34, 21, 26, 45, 38, 51, 42, 47.

Ex.2) Find median for the data 89, 71, 66, 53, 60, 75, 83, 59, 63, 49

Case-II) Discrete frequency distribution- let $X_1, X_2, X_3, \dots, X_n$ be the observations having frequencies $f_1, f_2, f_3, \dots, f_n$ respectively, it's Median is obtained as follows

- i) Arrange the values of variable in ascending order.
- ii) find l.c.f.

iii) Median = size of item having l.c.f. just $\geq \left(\frac{N+1}{2} \right)$

Ex.1) find median for the following data.

X : 14 16 17 19 23 27 29 32 35

F : 8 12 18 23 20 14 10 9 4

Ex.2) find median for the data given below.

X : 33 37 42 48 51 55 59 63 70

F : 11 16 24 27 31 29 23 12 8

Case-III) Continuous frequency distribution- let classes for the values of variable and their respective frequencies are given, its Median is obtained as follows.

i) Arrange the classes in ascending order.

ii) find l.c.f.

iii) find Median class-

Median class = class having l.c.f. just $\geq (N/2)$

$$\text{iv) Median} = L + \left(\frac{N/2 - cf}{f} \right) \cdot h$$

Where L = lower limit of median class

cf = l.c.f. of pre median class

f = frequency of median class

h = width of median class

Ex.1) Find median for the data given below.

Class : 15-25 25-35 35-45 45-55 55-65 65-75 75-85

Freq. : 7 16 23 31 27 18 11

Note – In case of inclusive classes, convert the classes into exclusive classes and then find median.

Ex.2) Find median for the data given below.

Class : 10-19 20-29 30-39 40-49 50-59 60-69 70-79

Freq. : 3 8 11 16 13 9 7

Merits and Demerits of Median

Merits:- i) it is simple to understand and easy to calculate.

ii) it is clearly and rigidly defined.

iii) it is not affected by extreme observations

iv) it is an actual value in the data

v) it is suitable for open-end classes.

vi) it can be determined graphically.

vii) it is applicable for qualitative data.

Demerits i) it is not based on all observations.

ii) it is not capable for further algebraic treatments.

iii) it has no sampling stability.

iv) Computations is based on arrangement of data.

Quartiles - We know that median divides the data into two equal parts, when data is arranged in ascending order of magnitude. Now suppose we want to divide the data in four equal parts, this can be done with the help of three points, these three points are known as quartiles.

Def. – Quartiles are defined as sizes of three points which divides the data into four equal parts, when data is arranged in ascending order of magnitude

$$\text{i.e. } Q_i = \text{size of } i\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item, } i=1,2,3$$

Note- i) The points Q_1 , Q_2 and Q_3 are called first, second and third quartiles respectively

ii) $Q_1 < Q_2 < Q_3$

iii) Median = Q_2

iv) Q_1 divides the data in the ratio 1:3

Q_2 divides the data in the ratio 1:1

Q_3 divides the data in the ratio 3:1

Def- Q_1 (first quartile) – it is defined as size of item which divides the data in the ratio 1:3, when

the data is arranged in ascending order of magnitude

$$\text{i.e. } Q_1 = \text{size of } 1\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

Def- Q_3 (third quartile) – it is defined as size of item which divides the data in the ratio 3:1, when

the data is arranged in ascending order of magnitude

$$\text{i.e. } Q_3 = \text{size of } 3\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

Computations of Quartiles.---

Case-I) Individual data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations on variable X , it's Quartiles are obtained as follows

i) Arrange the data in ascending order

$$\text{ii) } Q_i = \text{size of } i\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item, } i=1,2,3$$

Ex.1) Find Quartiles for the data 27, 44, 31, 36, 55, 48, 61, 52, 57.

Ex.2) Find Quartiles for the data 189, 171, 166, 153, 160, 175, 183, 159, 163, 149

Case-II) Discrete frequency distribution- let $X_1, X_2, X_3, \dots, X_n$ be the observations having frequencies $f_1, f_2, f_3, \dots, f_n$ respectively, it's Quartiles are obtained as follows

i) Arrange the values of variable in ascending order.

ii) find l.c.f.

$$\text{iii) } Q_i = \text{size of item having l.c.f. just } \geq i\left(\frac{N+1}{4}\right), i=1,2,3$$

Ex.1) find quartiles for the following data.

X : 14 16 17 19 23 27 29 32 35

F : 8 12 18 23 20 14 10 9 4

Ex.2) find quartiles for the data given below.

X : 33 37 42 48 51 55 59 63 70

F : 11 16 24 27 31 29 23 12 8

Case-III) Continuous frequency distribution- let classes for the values of variable and their respective frequencies are given, it's Quartiles are obtained as follows.

i) Arrange the classes in ascending order.

ii) find l.c.f.

iii) find Q_i^{th} class-

$$Q_i^{\text{th}} \text{ class} = \text{class having l.c.f. just } \geq i(N/4), i=1,2,3$$

$$iv) Q_i = L + \left(\frac{iN/4 - cf}{f} \right) \cdot h$$

Where L = lower limit of Q_i^{th} class

cf = l.c.f. of pre Q_i^{th} class

f = frequency of Q_i^{th} class.

h = width of Q_i^{th} class

Ex.1) Find quartiles for the data given below.

Class :	11-22	22-33	33-44	44-55	55-66	66-77	77-88
Freq. :	7	16	23	31	27	18	11

Note – In case of inclusive classes, convert the classes into exclusive classes and then find quartiles

Ex.2) Find quartiles for the data given below.

Class :	10-19	20-29	30-39	40-49	50-59	60-69	70-79
Freq. :	3	8	11	16	13	9	7

3) Mode Def- Mode is defined as size of item which is repeated most frequently in the data
OR size of item having maximum frequency.
i.e. Mode = size of item which occurs most

Computations of Mode----

Case-I) Individual data - let $X_1, X_2, X_3, \dots, X_n$ be the n observations on variable X , in this
case mode is the size of item which is repeated most
i.e. Mode = size of item which occurs most

Ex.1) Find median for the data 3, 4, 2, 2, 4, 3, 5, 4, 4.

Ex.2) Find median for the data 8, 9, 7, 6, 6, 5, 3, 6, 7, 5, 8, 3, 5, 9, 6, 3, 4, 9

Case-II) Discrete frequency distribution- let $X_1, X_2, X_3, \dots, X_n$ be the observations
having frequencies $f_1, f_2, f_3, \dots, f_n$ respectively, in this case

mode
is the size of item having maximum frequency
i.e. Mode = size of item having maximum frequency

Ex.1) find median for the following data.

X :	14	16	17	19	23	27	29	32	35
F :	8	12	18	23	20	14	10	9	4

Ex.2) find median for the data given below.

X :	33	37	42	48	51	55	59	63	70
F :	11	16	24	27	31	29	23	12	8

Case-III) Continuous frequency distribution- let classes for the values of variable and their respective frequencies are given , in this case mode is obtained as follows.

i) find model class-

model class = class having maximum frequency

$$\text{ii) Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h$$

Where L = lower limit of model class

f_1 = frequency of model class

f_0 = frequency of pre model class

f_2 = frequency of post model class

h = width of model class

Merits and Demerits of Mode

- Merits:-**
- i) it is simple to understand and easy to calculate.
 - ii) it can be determined graphically.
 - iii) it is not affected by extreme observations
 - iv) it is an actual value in the data
 - v) it is suitable for open-end classes.
 - vi) it is applicable for qualitative data.

- Demerits**
- i) it is not based on all observations.
 - ii) it is not capable for further algebraic treatments.
 - iii) it has no sampling stability.
 - iv) Computations is based on arrangement of data.
 - v) it is not rigidly defined.

Ex.1) Find mode for the data given below.

Class :	115-125	125-135	135-145	145-155	155-165	165-175	175-185
Freq. :	7	16	23	31	27	18	11

Note – In case of inclusive classes, convert the classes into exclusive classes and then find median.

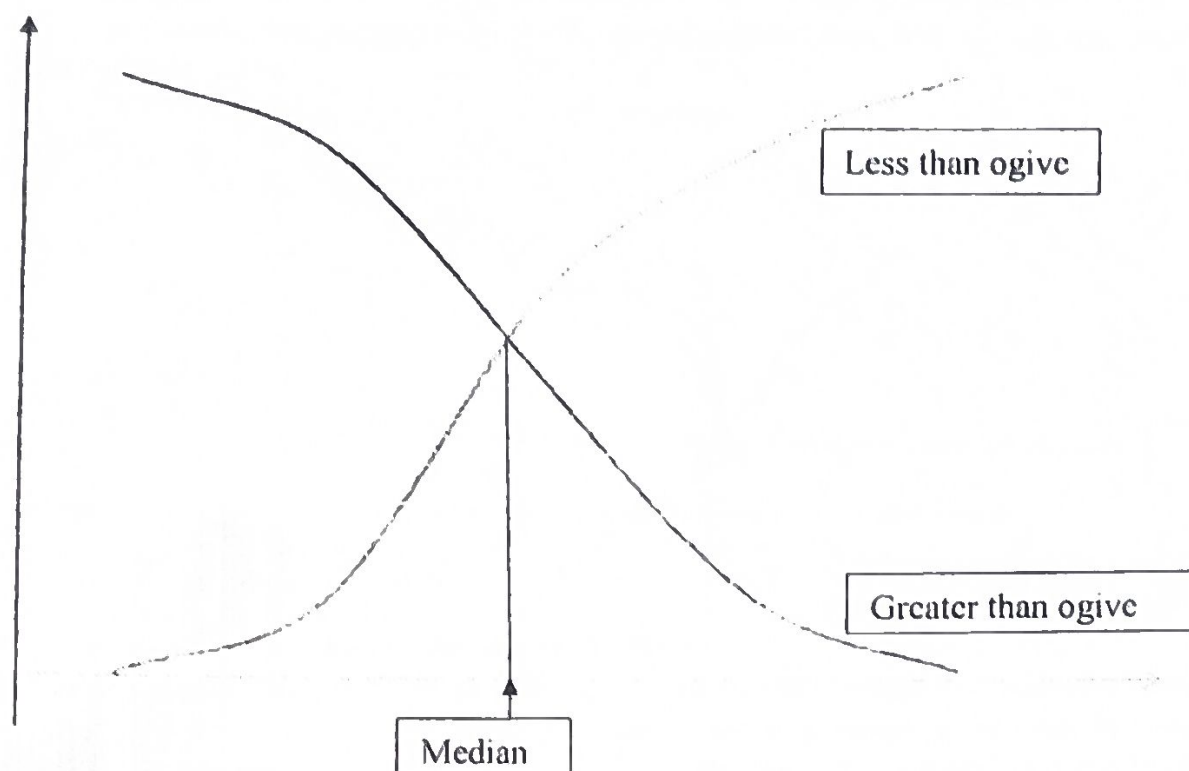
Ex.2) Find mode for the data given below.

Class :	10-19	20-29	30-39	40-49	50-59	60-69	70-79
Freq. :	3	8	11	16	13	9	7

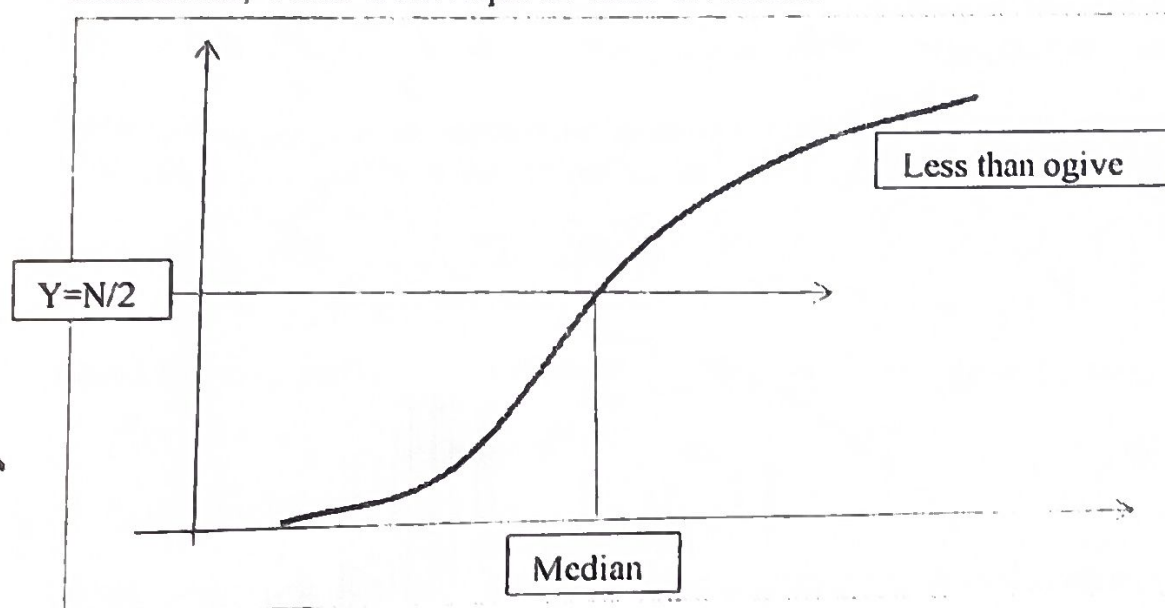
Determination of median Graphically –

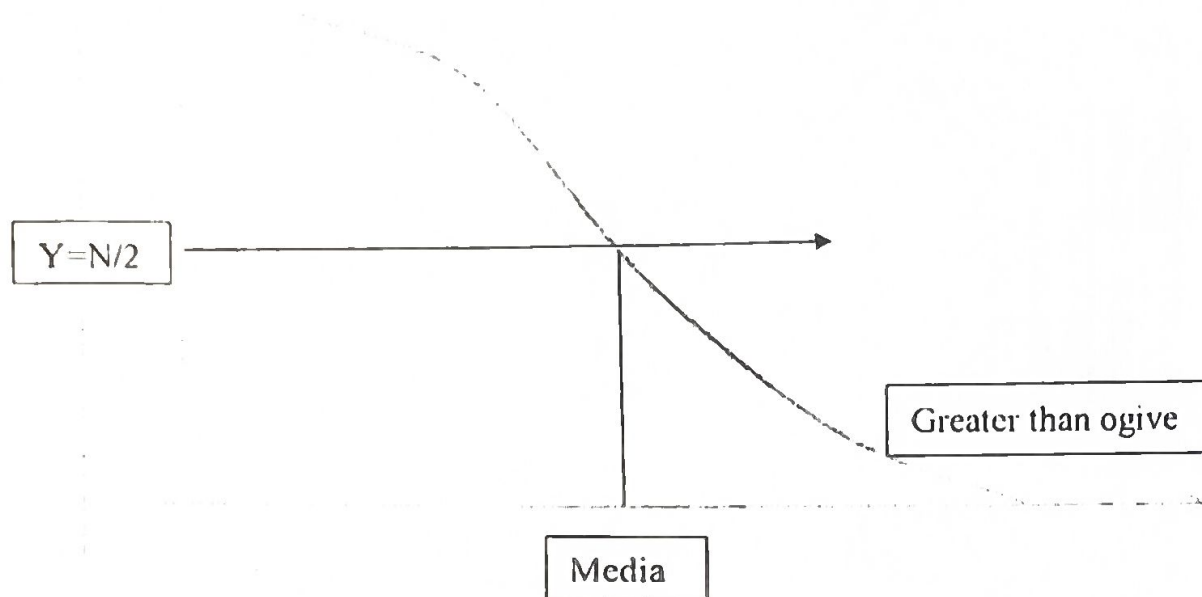
Median is the size of item which at center, when data is arranged in ascending or descending order of magnitude. So Median is obtained graphically by using ogives. There are two ways by which median is obtained

- i) By drawing both ogives – Draw both less than and greater than ogives on the same graph paper. These ogives intersect each other, from the point of intersection drop a perpendicular to X-axis and read the X- co-ordinates of the point of intersection, which is the required value of Median.



ii) By drawing one of the ogive- Draw only less than or greater than ogive. Draw a parallel line to X-axis at point $Y = N/2$, this line intersects the ogive drawn already, from the point of intersection drop a perpendicular to X-axis and read the X- co-ordinates of the point of intersection, which is the required value of Median.

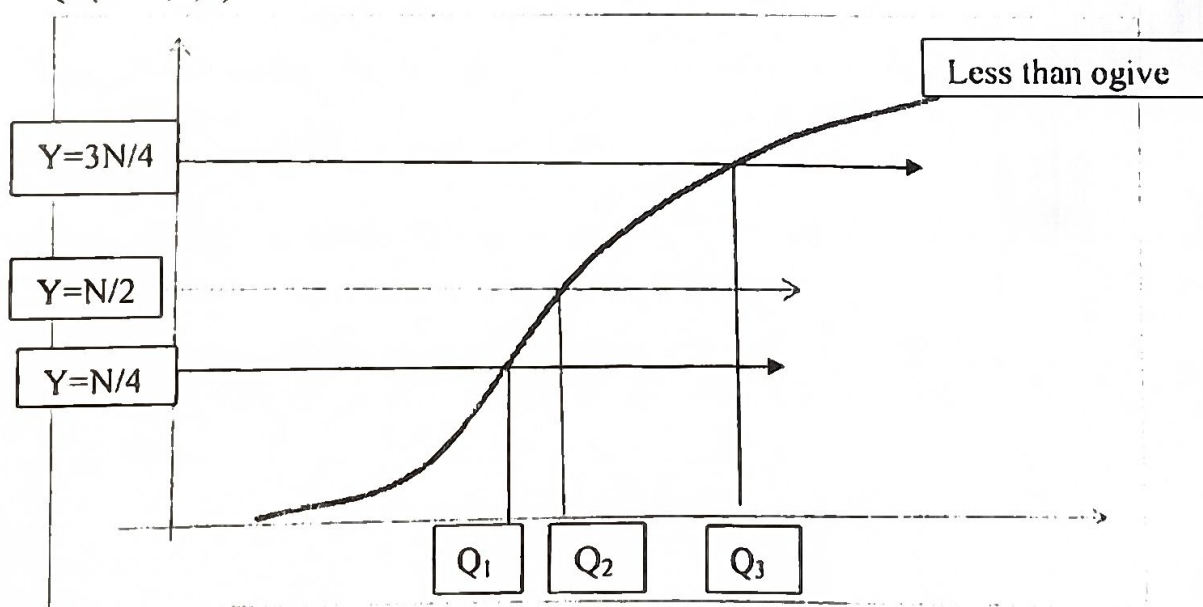




Determination of Quartiles Graphically –

Quartiles is the sizes of three items which divides the data in four equal parts, when data is arranged in ascending order of magnitude. So to quartiles are determined by using less than ogive Hence first draw less than ogive to represent the given data. Draw parallel line to X-axis at point $Y = iN/4$

($i = 1, 2, 3$). This line intersects the ogive, from the point of intersection drop a perpendicular to X-axis and read the X- co-ordinate of the point of intersection, which is the required value of Q_i ($i = 1, 2, 3$)

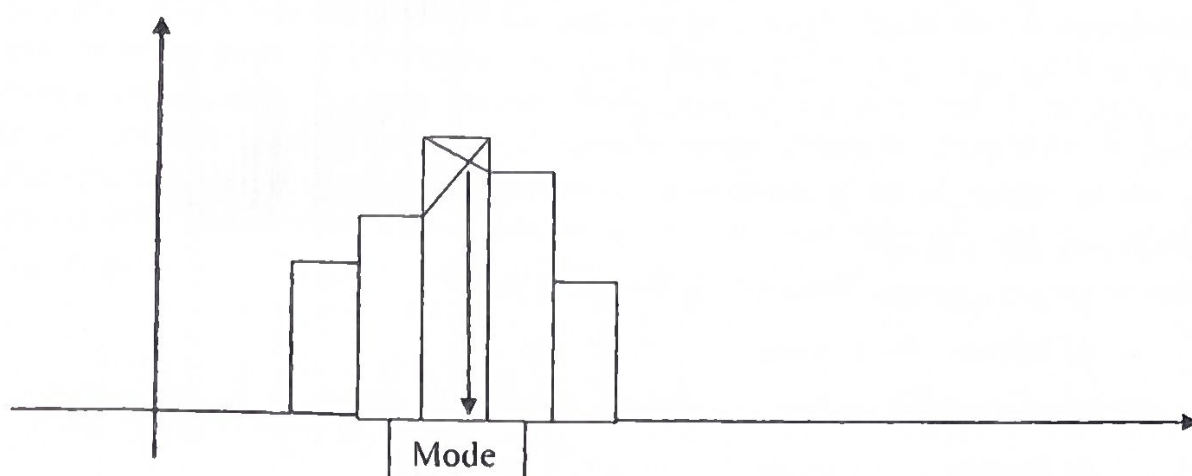


Determination of Mode Graphically –

Mode is determined by using histogram. So draw a histogram to represent the given data. Since mode is the size of item having maximum frequency, therefore it lies in the class having a rectangle with maximum height. Select a rectangle having maximum height and go through following steps.

- i) connect right hand top corner of highest rectangle to right hand top corner of previous rectangle by segment of straight line.
- ii) connect left hand top corner of highest rectangle to left hand top corner of post rectangle by segment of straight line.

iii) the segments drawn in above two steps are intersects each other. drop a perpendicular to X-axis from the point of intersection and read the X-co-ordinate of the point of intersection, which is the required value of mode.



@Empirical relation between Mean, Median and Mode

From the experience with large number unimodal frequency distribution it is observed that the median lies between mean and mode. Based on this vast experience a relation between 3 measures of central tendency has been established and it is known as empirical relation between mean, median and mode

i) For a moderately asymmetric unimodal distribution the following relation called empirical holds between mean, median and mode.

$$(\text{Mean} - \text{Mode}) = 3(\text{Mean} - \text{Median})$$

This relation does not have any proof or mathematical base.

ii) For symmetric distribution- $\text{Mean} = \text{Median} = \text{Mode}$.

Examples –

1) Find missing frequency of a class 54 – 72 from the following data if median is 65

class	0 - 18	18 - 36	36 - 54	54 - 72	72 - 90	90 - 108	108 - 126
frequency	4	6	12	--	14	9	3

2) Find missing frequency of a class 40 – 55 from the following data if mode is 51

class	10 - 25	25 - 40	40 - 55	55 - 70	70 - 85
frequency	12	17	--	24	11

3) Find missing frequency of a value 13 from the following data if mean is 15.10

X	10	11	12	13	14	15
frequency	5	7	10	--	8	3

4) Find missing frequency of a class 35 – 45 from the following data if mean 41.60

class	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65
frequency	5	10	--	14	7

5) The mean and mode of a frequency distribution are 40 and 30 respectively. Find median.

6) Find $\bar{X}$ if $n = 25$ and $\sum u = 144$, where $u = (x - 24)/5$

7) Following data shows distribution of 132 items, Find missing frequencies if median is 34.5

class	8-16	16-24	24-32	32-40	40-48	48-56	56-64
frequency	11	--	27	--	21	16	7

7) Choose most correct alternative

- It is capable of further algebraic treatment is the merit of.....
a) A.M. b) Median c) Mode d) all of these
- measure of central tendency is affected by extreme observations.
a) A.M. b) Median c) Mode d) all of these
- measure of central tendency is determined from frequency polygon .
a) A.M. b) Median c) Mode d) none of these
- Quartiles are determined graphically using.....
a) histogram b) less than ogive c) frequency curve d) all of these
- The class having maximum frequency is known as.....
a) median class b) model class c) quartile class d) none of these.
- The A.M. of 10 values is 25, if each observation is increased by 3 and then doubled, the A.M becomes.....
a) 25 b) 56 c) 53 d) none of these.
- If each value is increased by 10, then mode becomes.....
a) as it is b) increased by 10 c) decreased by 10 d) none of these
- The mean and median of a distribution are 10 and 15 respectively, the value of mode becomes.....
a) 25 b) 15 c) 20 d) none of these
- measure of central tendency is not suitable for open-end classes.
a) A.M. b) Median c) Mode d) all of these
- If all the observations in the data are equal, then.....
a) mean=median=mode b) mean>median>mode
c) mean<median<mode d) all of these
- The algebraic sum of deviations of observations taken from their A.M. is ----- always.
a) zero b) any +ve number c) any -ve number d) none of these
- To determine ----- measure of central tendency, data should be arranged in ascending or descending order.
a) A.M. b) Median c) Mode d) all of these
- First quartile divides the data in the ratio ----- when data is arranged in ascending order.
a) 1:3. b) 1:4 c) 3:1 d) 1:2
- Quartiles are the ----- number of points that divides the data in to 4 equal parts when data is arranged in ascending order.
a) 3 b) 4 c) 2 d) none of these
- Measure of central tendency are affected by change in -----
a) origin b) scale c) origin and scale both d) none of these